Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	SWTID-2026-3296
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability: This document evaluates the documented code structure, readability, modularity, and reusability of the ComicCraft application based on the supplied project document.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Not specified	Source document does not provide enough code samples to verify indentation.
2	Proper File Structure	Files and folders are logically organized	Yes	The project is separated into modules such as routes.py, gemini_flash.py, gemini_pro.py, image_generator.py, layout_builder.py and exporters.py.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Documented names such as story prompt, character name, setting, tone, art style, layout and panel are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions include generate_outline(), generate_story(), generate_image(), build_comic_layout() and save_pdf().
5	Code Comments	Inline and block comments explain logic	Not specified	The supplied project document describes functions but does not provide enough source code comments to verify this.
6	Modular Design	Code is split into reusable functions/modules	Yes	Core responsibilities are separated into dedicated modules and functions.
7	No Redundant Code	Duplicate or unused code is removed	Not specified	Redundancy cannot be verified from the project description alone.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	The document states that FastAPI HTTPException and try-except blocks are used for unexpected failures such as API or JSON errors.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	generate_outline() / gemini_flash.py	Python / Gemini Flash	Creates the structured 5-panel outline used in the comic generation workflow.	High
2	generate_story() / gemini_pro.py	Python / Gemini Pro	Expands panel outlines into narration and character dialogue.	High
3	generate_image() / image_generator.py	Python / Stable Diffusion / Diffusers	Generates panel illustrations from image prompts and is also used by the /test-image route.	High
4	build_comic_layout() / layout_builder.py	Python	Matches generated images with panel stories and structures the final layout.	High
5	save_pdf() / exporters.py	Python / FPDF	Compiles comic images and narration into the downloadable PDF export.	High
6	Jinja2 Templates	HTML / CSS / Jinja2	index.html, comic_preview.html and export_success.html are rendered by FastAPI routes.	High
7	FastAPI routes	Python / FastAPI	/generate and /generate-comic/json use the same core generation workflow for form and JSON requests.	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	N/A	The supplied project document describes a modular structure, but a full source-code review was not supplied.
Readability	N/A	Descriptive function/module names are documented; complete source formatting was not available for assessment.
Reusability	N/A	The project explicitly separates reusable functions/modules and uses them across routes.
Documentation / Comments	N/A	Project documentation is detailed, but inline source-code comments cannot be verified from the supplied document.
Overall Score	N/A	No numeric score is assigned because the assessment is based on project documentation rather than a full code inspection.

Source Note: This document is prepared from the supplied ComicCraft project document and follows the supplied Code-Layout, Readability and Reusability template. Items not supported by the source are explicitly marked as not specified or N/A.